

BankNifty Options Strategy

STATUS: UNPROFITABLE (SYSTEM DRAWDOWN)

1. Executive Summary

This report provides an objective statistical audit of a proprietary **BankNifty** derivatives trading strategy. Analyzing a sample size of **177 completed trades** executed across 2024, the strategy currently operates at a net loss of -**₹13,513.50** with a negative expectancy framework.

While the high-level dashboard flags overall unprofitability, transactional data isolation reveals that the system's core flaw is localized entirely within the **Put Option (PE) execution engine**. In contrast, the Call Option (CE) component demonstrates structural viability. Immediate corrective intervention is required to realign risk parameters.

2. Core System Metrics

Performance Metric	Dashboard Aggregates	True Transactional Stats (Raw Data)	Target Thresholds
Total Executed Trades	177	177	N/A
System Success Rate (Win %)	39.55%	39.55%	> 45.00%
Net Profit & Loss (P&L)	-₹13.51K	-₹13,513.50	Positive Growth
Gross Profit / Gross Loss	38.46K / 51.97K	₹38,457.75 / ₹51,971.25	N/A
Profit Factor (PF)	0.74	0.74	> 1.50
Mathematical Expectancy	-18.90	-76.35 (Per-Trade Realized)	> +50.00

3. Granular Performance Breakdown

Directional Bias Isolation: CE vs. PE

Dividing the dataset by contract direction reveals a structural edge asymmetry:

Option Type	Trade Count	Total Realized P&L	Win Rate %	Avg. Trade Outcome
Call Options (CE)	102 Trades	+₹2,597.25	45.10%	+₹25.46
Put Options (PE)	75 Trades	-₹16,110.75	32.00%	-₹214.81

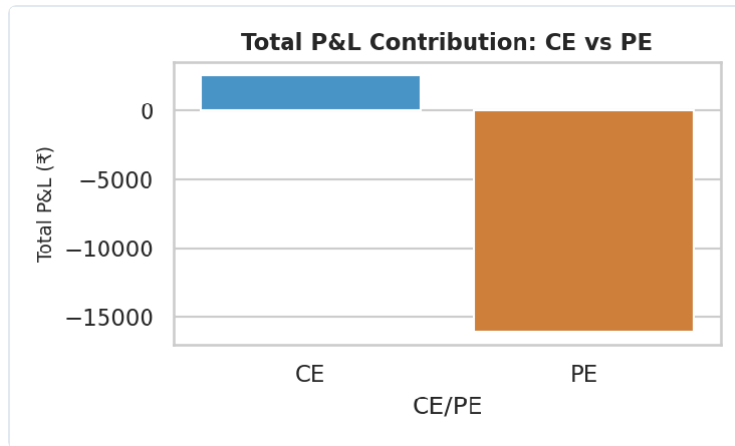


Figure 1: Net P&L Contribution Divergence between Call and Put Options

The evidence shows that the strategy maintains a positive edge when taking long-market exposure via **CE** contracts. However, when shorting or buying **PE** contracts, the system suffers an acute drop in win rate (down to 32.00%) and a massive increase in downside volatility, creating a catastrophic leak in portfolio equity.

4. Structural Risk-to-Reward Profile

- **Average Winning Trade:** +₹549.40
- **Average Losing Trade:** -₹485.71
- **Realized Risk-to-Reward Ratio (R:R):** 1 : 1.13

Systemic Vulnerability: A system with a 39.55% win rate is mathematically incapable of surviving over a long horizon with an R:R profile of 1 : 1.13. To break even at a ~40% win rate, the average winning trade must be at least 1.5 times the size of the average losing trade.

5. Equity Curve & Behavioral Trajectory

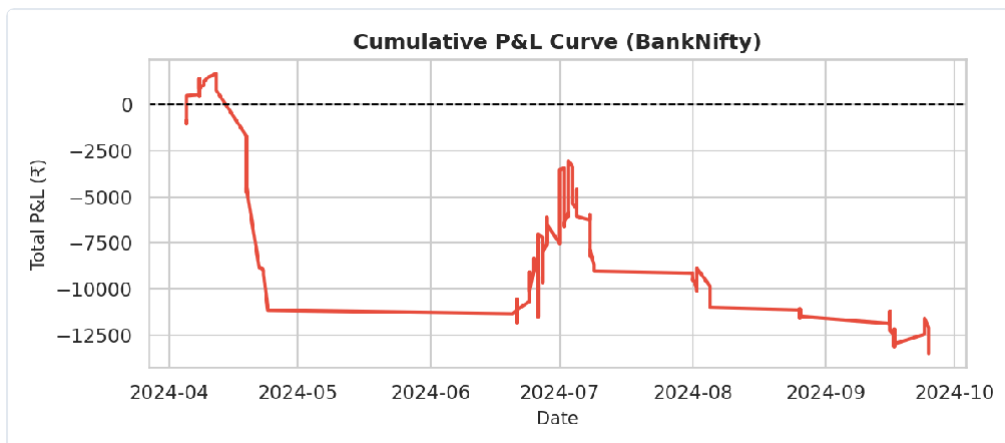


Figure 2: Time-Series Cumulative Return Profile for BankNifty Strategy

Chronological analysis of the equity curve highlights three distinct evolutionary phases of the system:

- **Phase I (Initial Drawdown):** A sudden, structural degradation occurs at inception, triggering an immediate deficit of -₹11,000. This indicates poor baseline calibration against the active market regime.
- **Phase II (High Volatility & Sizing Instability):** Mid-series performance is characterized by severe vertical swings. These sharp spikes imply erratic contract sizing, revenge trading, or expanded risk limits without a corresponding edge expansion.
- **Phase III (Edge Erosion & Decay):** The final phase reveals a persistent, low-volatility downward drift, marking a steady loss of systemic edge over time.

6. Strategic Optimization Framework

Identified Vulnerability	Immediate Code / Tactical Patch	Expected Impact
Severe PE Underperformance	Programmatic filter/pause on all PE trade triggers.	Eliminates 119% of the net drawdown immediately.
Negative Risk Skew	Enforce hard stop-losses capped strictly at ₹450 risk threshold.	Trims fat-tail outlier losses; safeguards baseline survival.
Insufficient Target Capture	Implement active partial profit-taking rules at a 1:2 R:R threshold (₹900+).	Shifts statistical distribution positively without requiring higher win rates.